Robustness across conditions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Robustness across conditions before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Inspecting failures and anomalies

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Reproducible environment and records

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Predefining an analysis plan

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Robustness across conditions trace and record the first step you can explain without help.

Use W2: Trace the worked case for Robustness across conditions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Robustness across conditions explanation against the supplied facts before discussing it.

Use the worked case for Robustness across conditions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Baseline/candidate on conditionA:6/8, 7/8; conditionB:5/8, 4/8. Compare paired rates.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Robustness across conditions facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 Trace the worked case

> Trace the worked case. Apply the supplied Robustness across conditions facts and keep your reasoning.

Method passes9/10 on current sources and5/10 on outdated sources. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Robustness across conditions facts and keep your reasoning.

A report omits this qualification: Robustness is bounded by the tested variations and independent **evidence** policy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Robustness across conditions facts and keep your reasoning.

Baseline/candidate on conditionA:6/8, 7/8; conditionB:5/8, 4/8. Compare paired rates.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Robustness across conditions facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Robustness across conditions to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled robustness across conditions **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of robustness across conditions.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Robustness across conditions

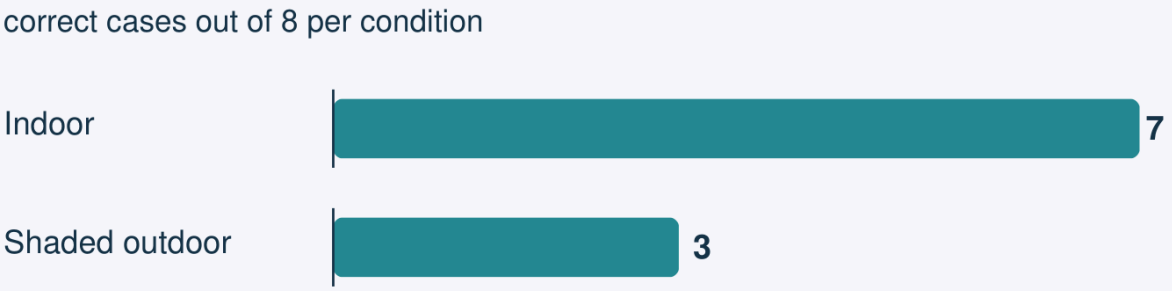
> Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases. Explain the labelled step before checking the changed condition.

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.

Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.

AI ACADEMY / AI-7/27

Fixed model across conditions



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/27

Robustness across conditions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name both slices

Each condition has eight labelled cases.



2. Compute rates

Indoor is 87.5%; outdoor is 37.5%.



3. Bound robustness

The large observed gap motivates investigation rather than claiming condition independence.

CHANGE ONE THING

Report only the combined 10/16=62.5%.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The large observed gap motivates investigation rather than claiming condition independence.

1. Name both slices

2. Compute rates

3. Bound robustness

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Bound robustness The large observed gap motivates investigation rather than claiming condition independence.

Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Report only the combined 10/16=62.5%. Which conclusion is supported by the supplied facts?

1. The arithmetic is correct, but the condition gap becomes hidden.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The arithmetic is correct, but the condition gap becomes hidden. Combining records discards the slice distinction relevant to the deployment goal.

Repair a missing building block

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the **mechanism** and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10.

What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

Review Reproducible environment and records (ai-7/24). Goal: Explain the **mechanism** and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9.

Which artefacts must accompany the **output** so another team can identify the exact run?. Return to this week and explain what you changed.

Review Predefining an analysis plan (ai-7/21). Goal: Explain the **mechanism** and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule.

Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

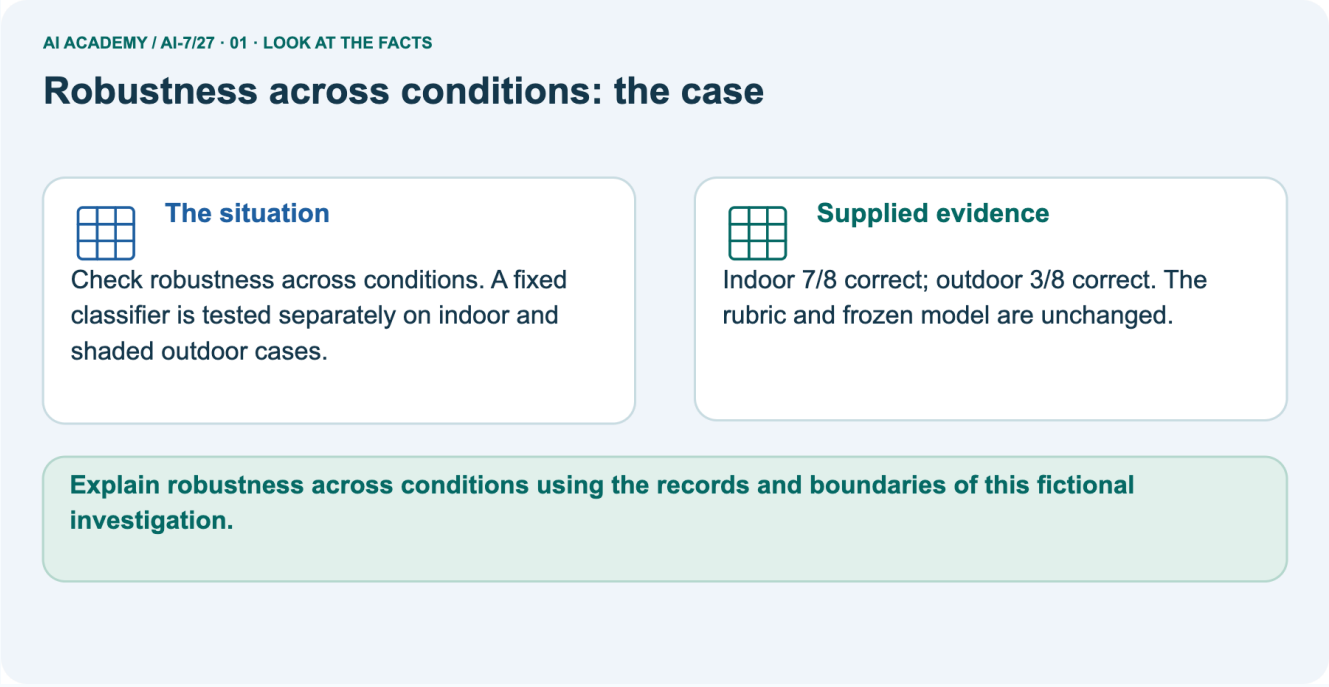
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Robustness across conditions picture from its labels, then record a relevant limit.

01. Robustness across conditions: the case



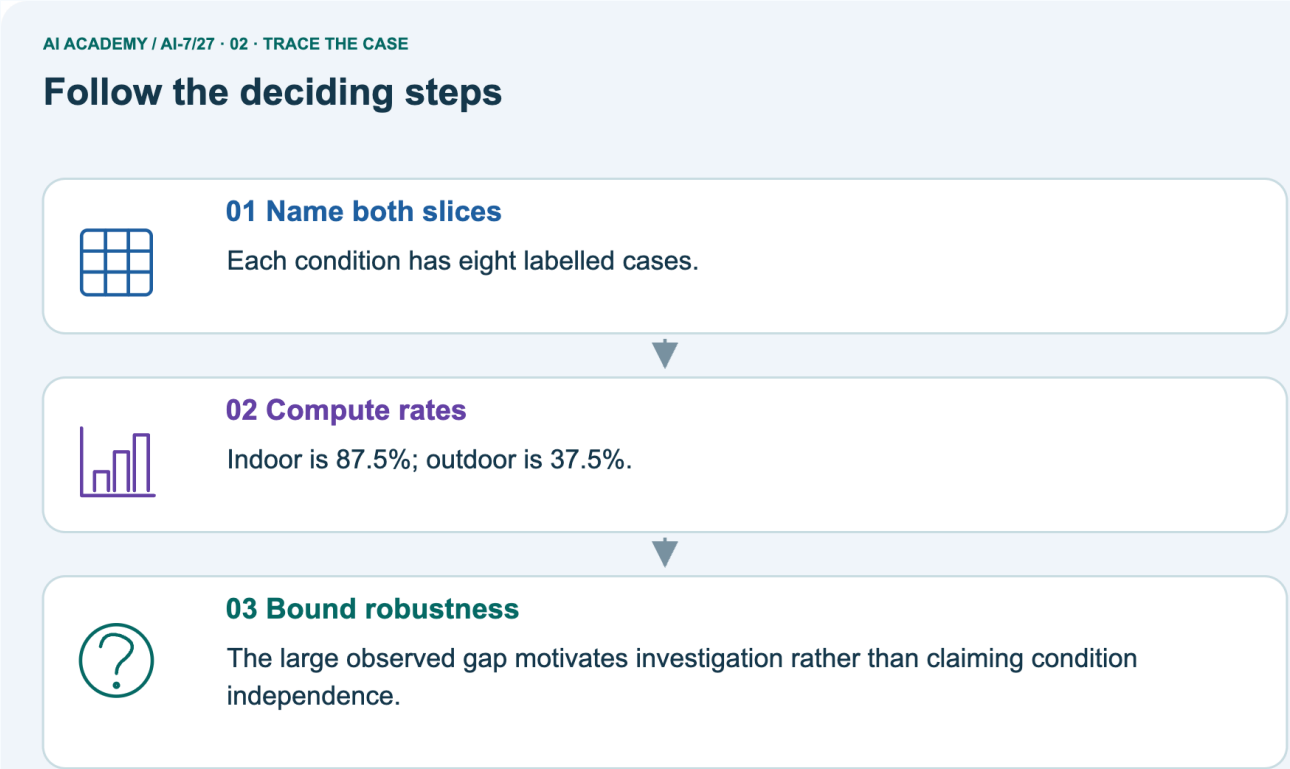
Follow the labelled picture with the matching explanation. Open the vector diagram

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases. Supplied facts: Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



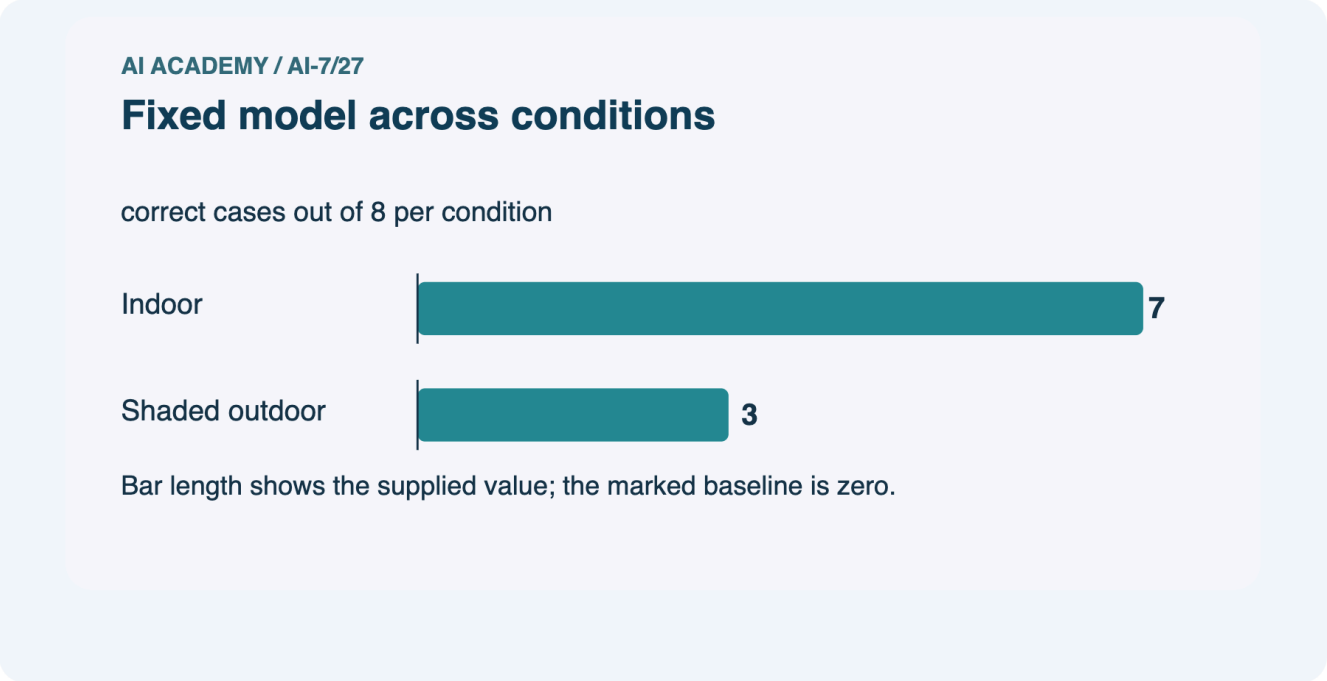
Follow the labelled picture with the matching explanation. Open the vector diagram

Name both slices: Each condition has eight labelled cases. → Compute rates: Indoor is 87.5%; outdoor is 37.5%. → Bound robustness: The large observed gap motivates investigation rather than claiming condition independence.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Fixed model across conditions



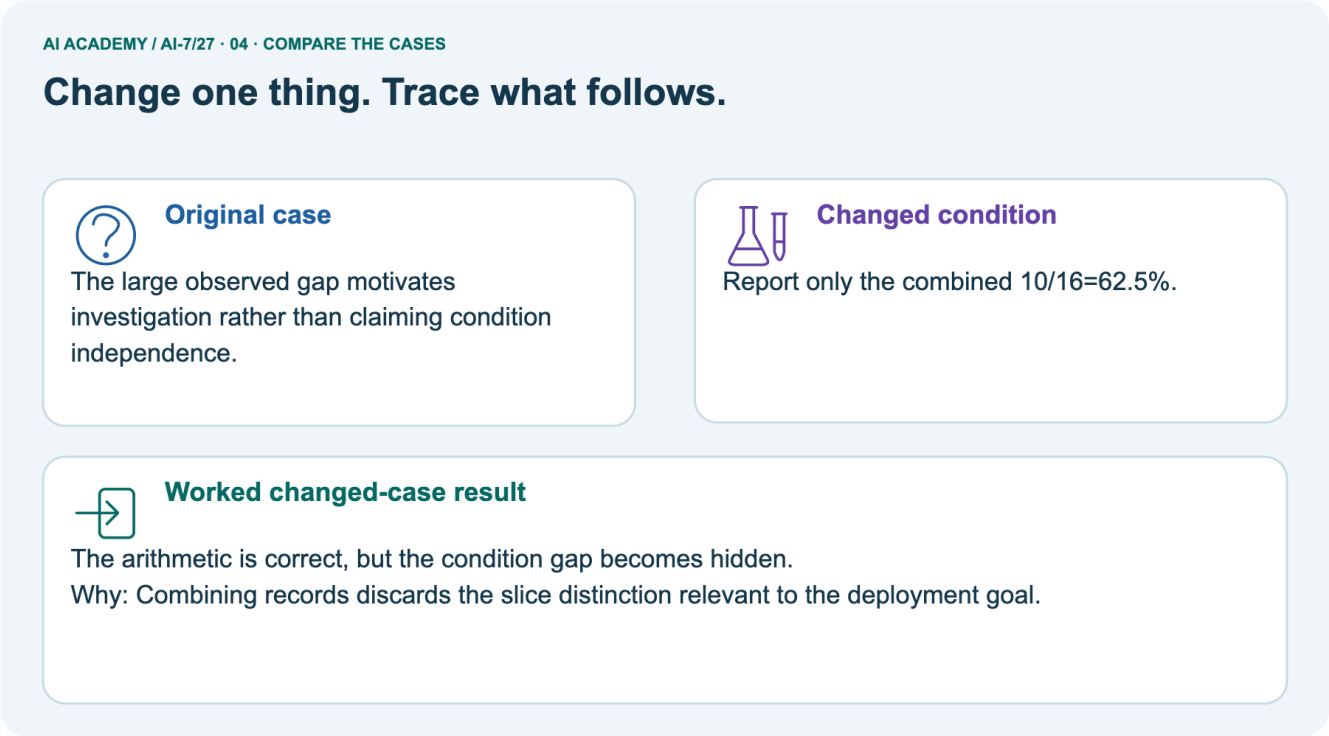
Follow the labelled picture with the matching explanation. Open the vector diagram

Fixed model across conditions. Units: correct cases out of 8 per condition. Indoor: 7. Shaded outdoor: 3.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

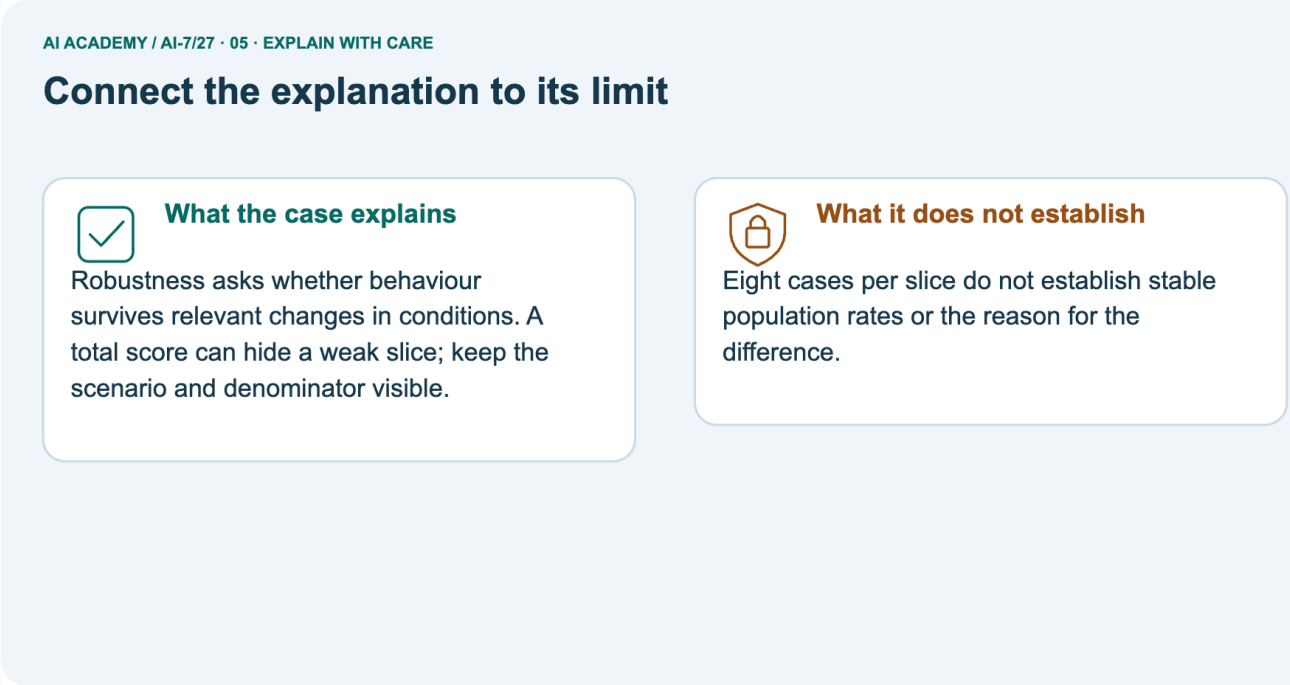
Original case: The large observed gap motivates investigation rather than claiming condition independence. Changed condition: Report only the combined 10/16=62.5%.

Expected result: The arithmetic is correct, but the condition gap becomes hidden. Reason: Combining records discards the slice distinction relevant to the deployment goal.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

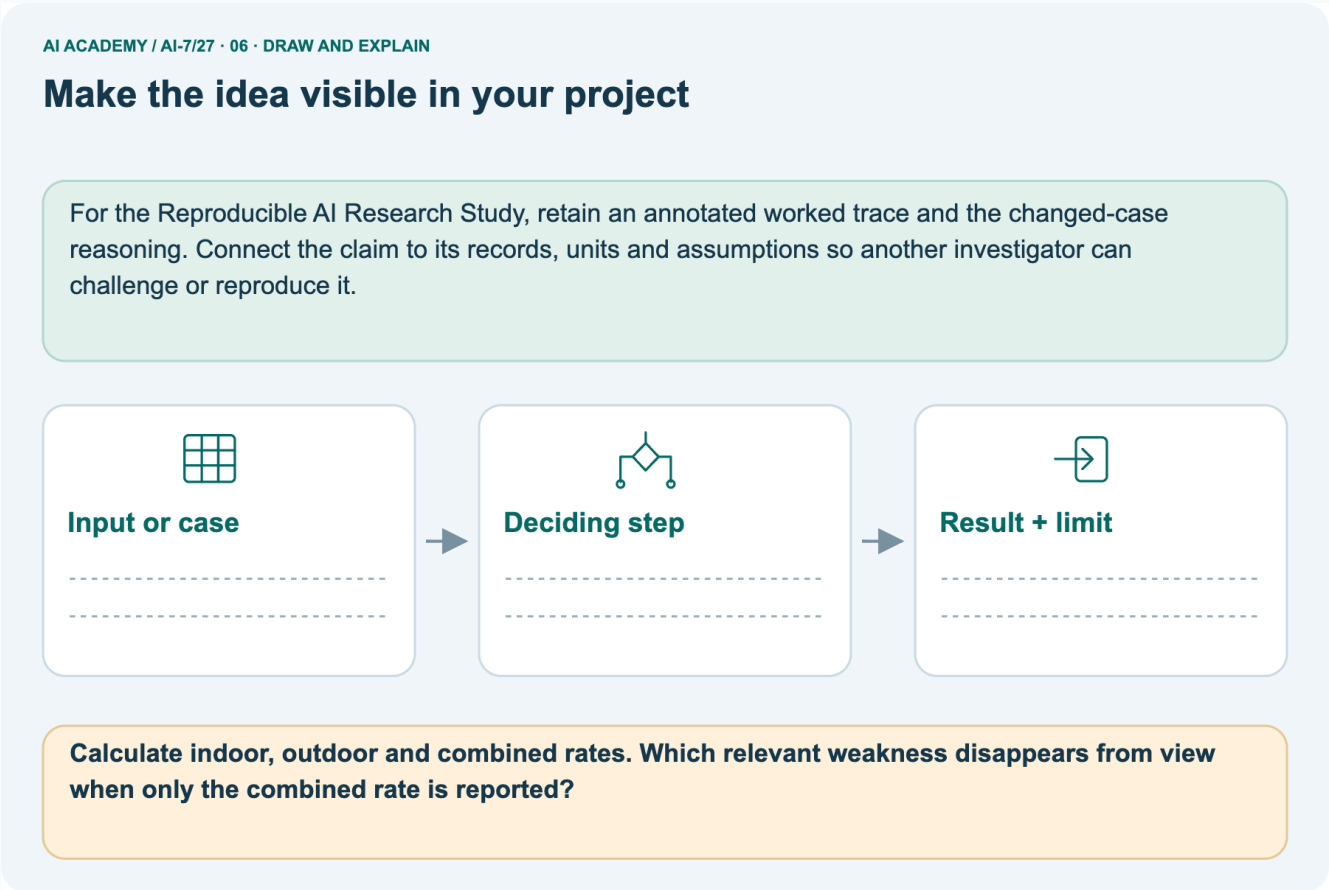
Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

Limit: Eight cases per slice do not establish stable population rates or the reason for the difference.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.